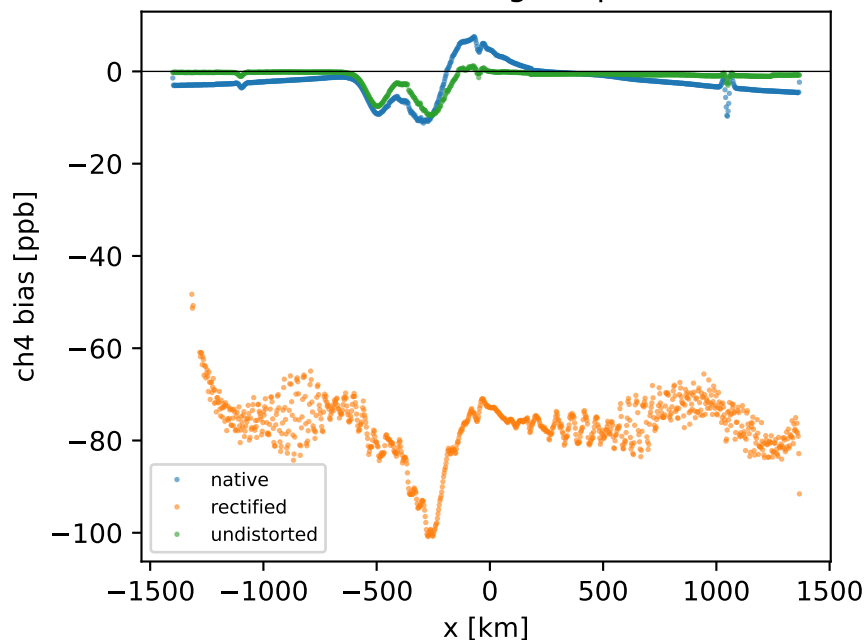
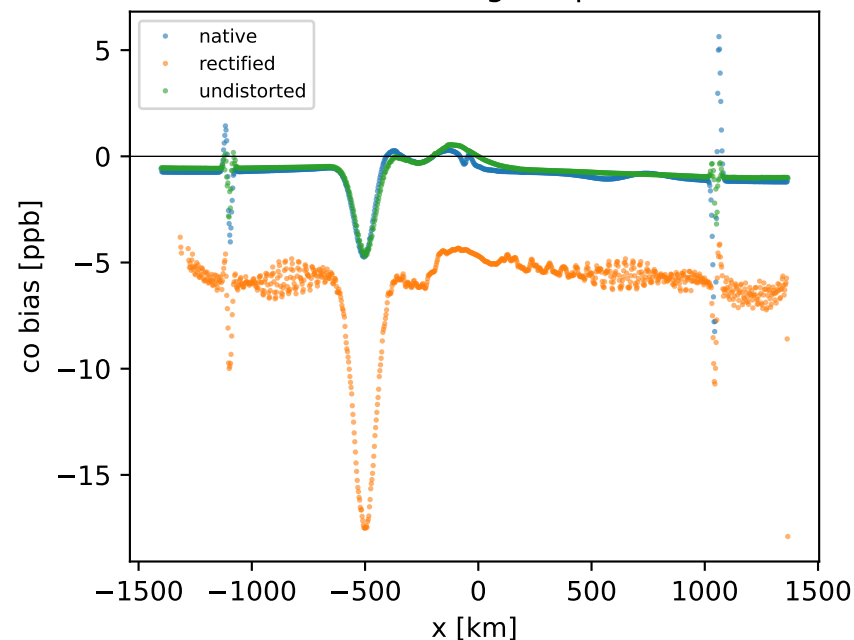


Joint FPA3 -- realistic, no noise (native/rectified/undistorted)

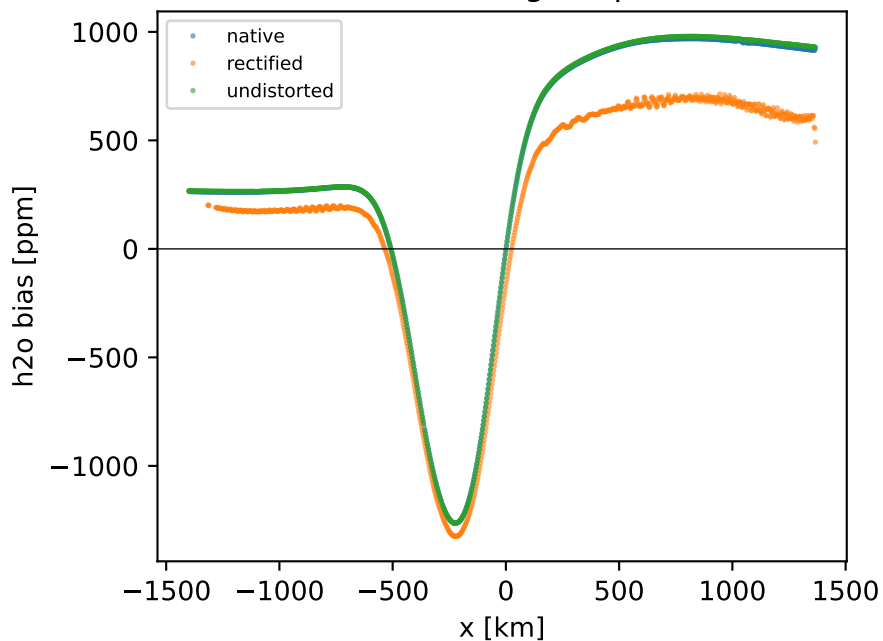
ch4 bias vs along-slit position



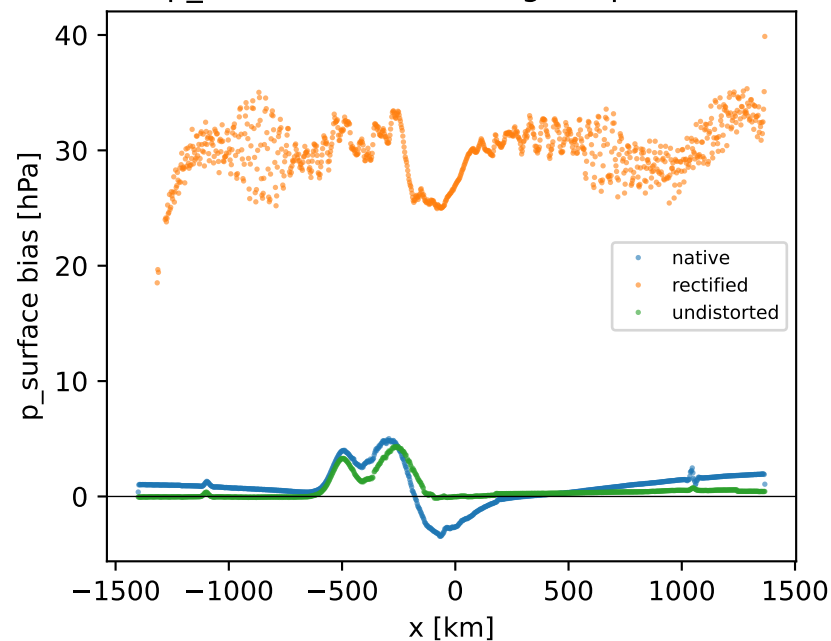
co bias vs along-slit position



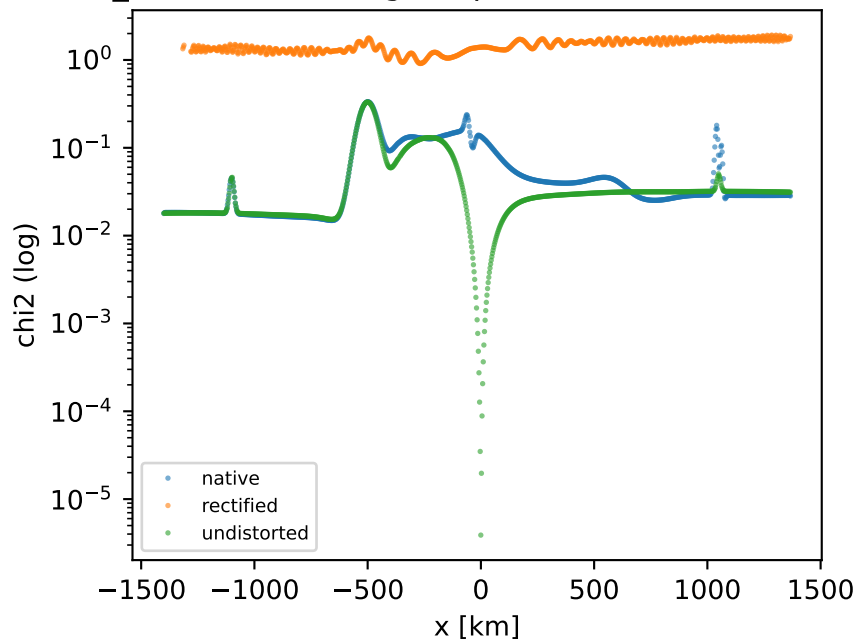
h2o bias vs along-slit position



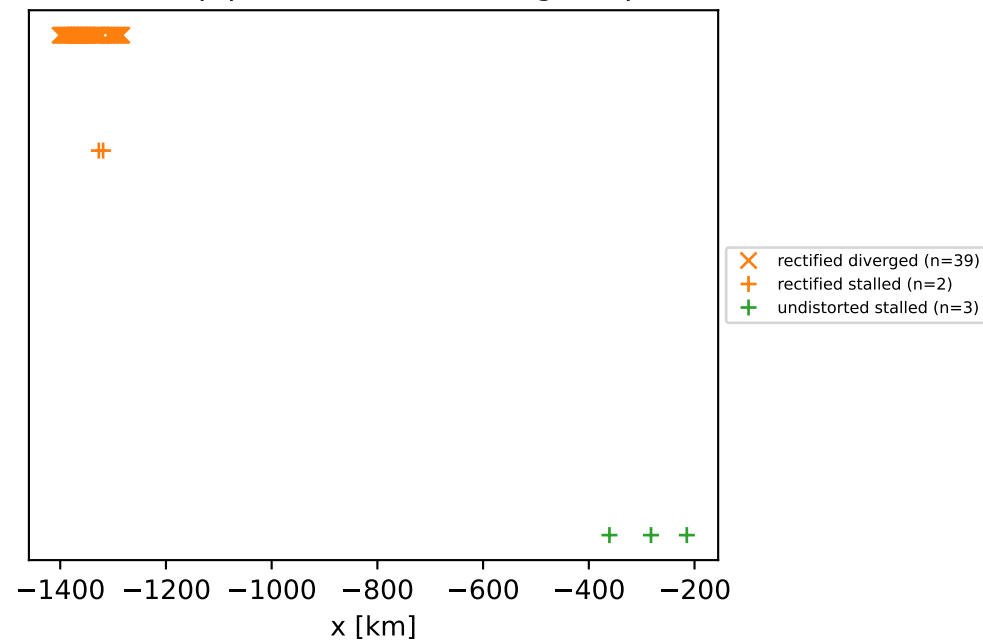
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

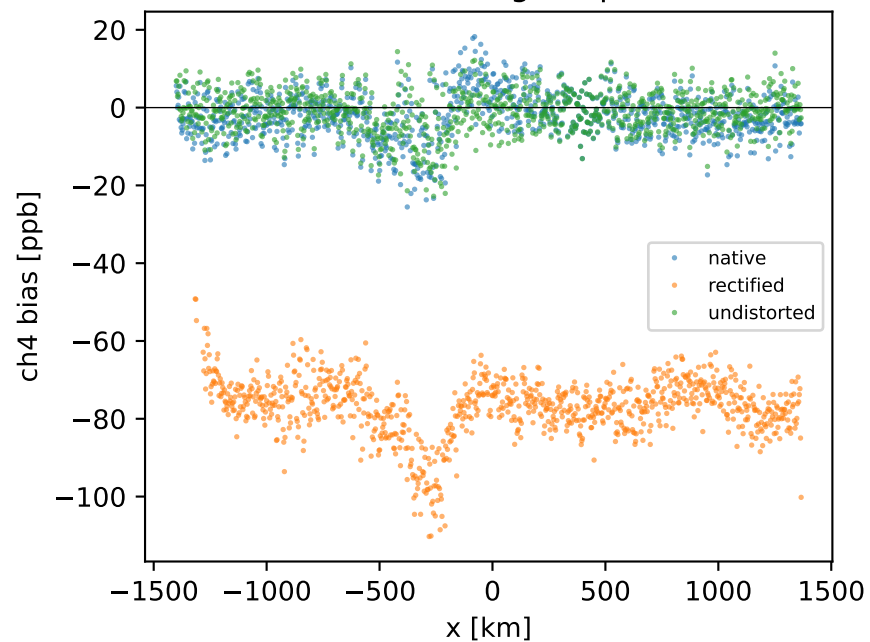


Where each pipeline fails, vs along-slit position

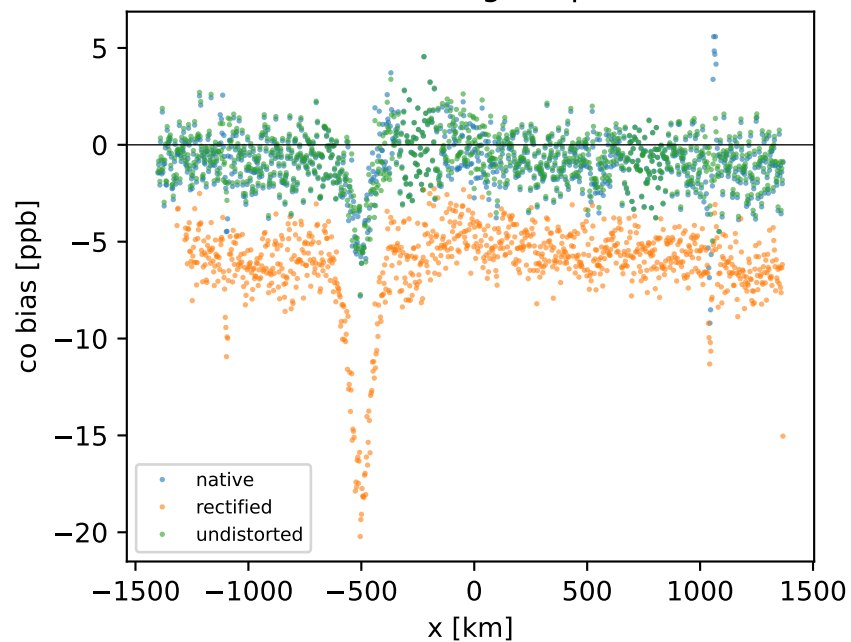


Joint FPA3 -- realistic, noise (native/rectified/undistorted)

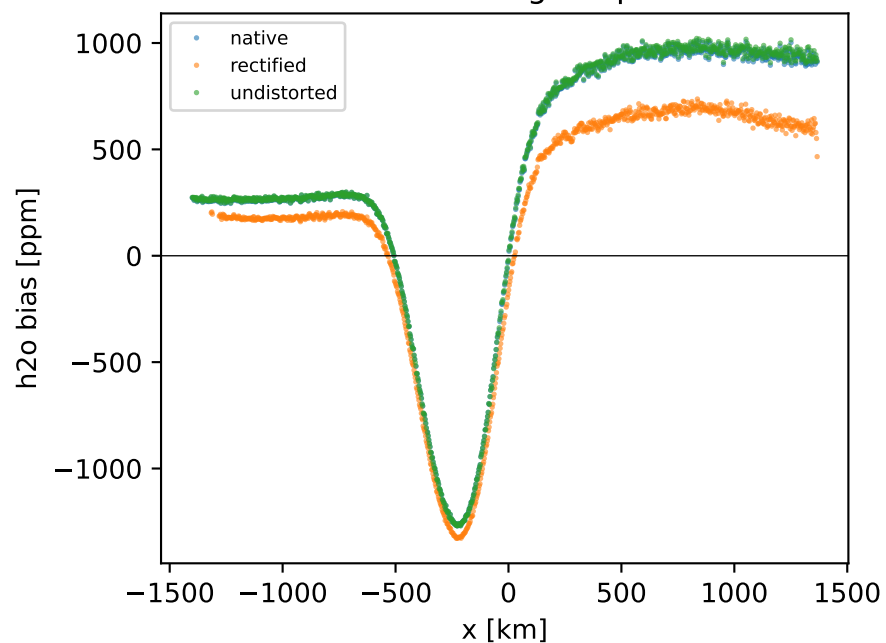
ch4 bias vs along-slit position



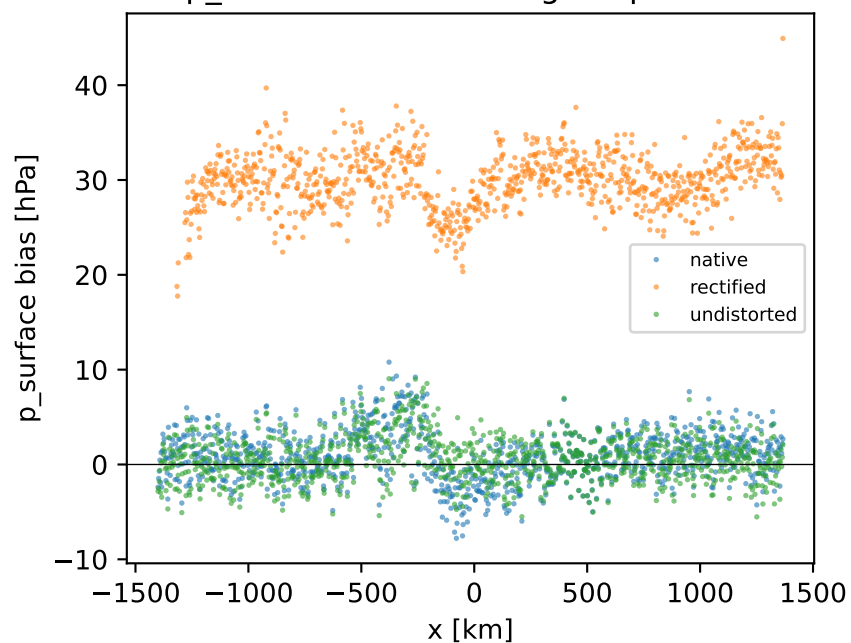
co bias vs along-slit position



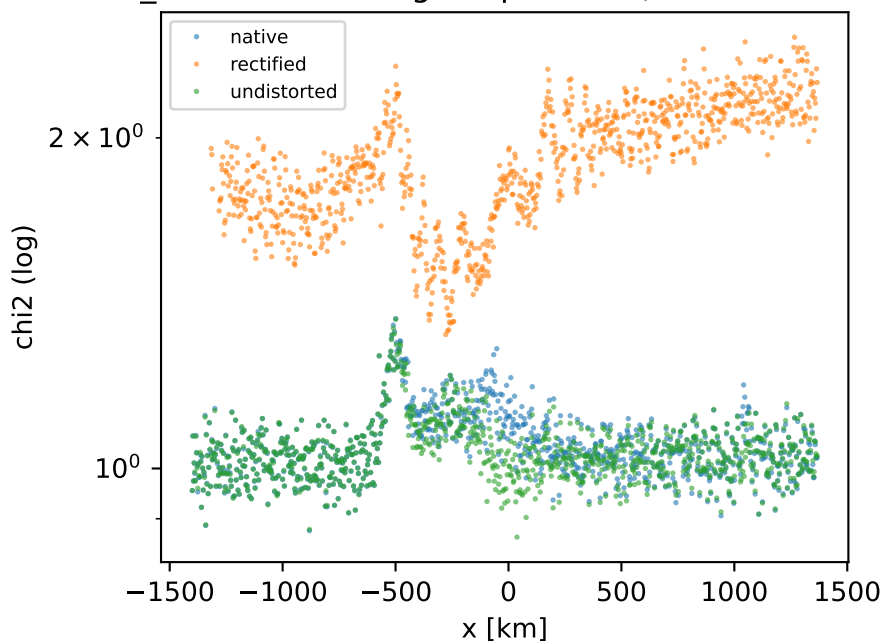
h2o bias vs along-slit position



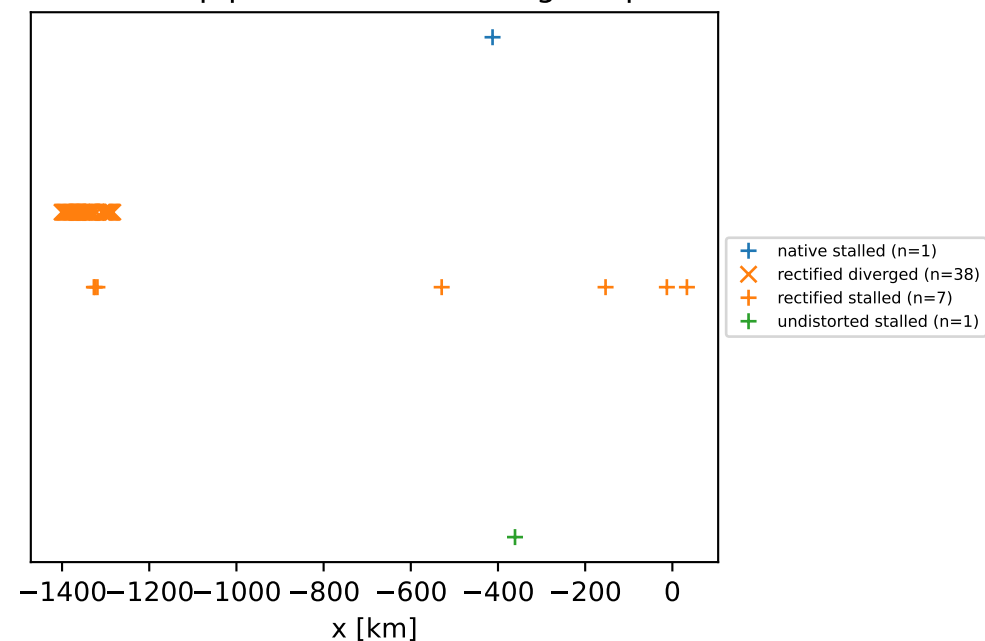
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

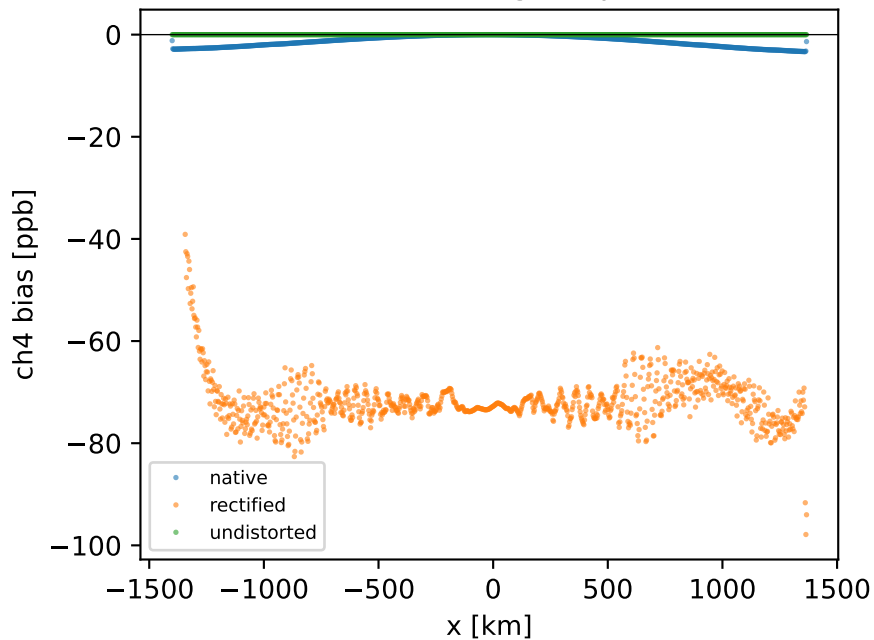


Where each pipeline fails, vs along-slit position

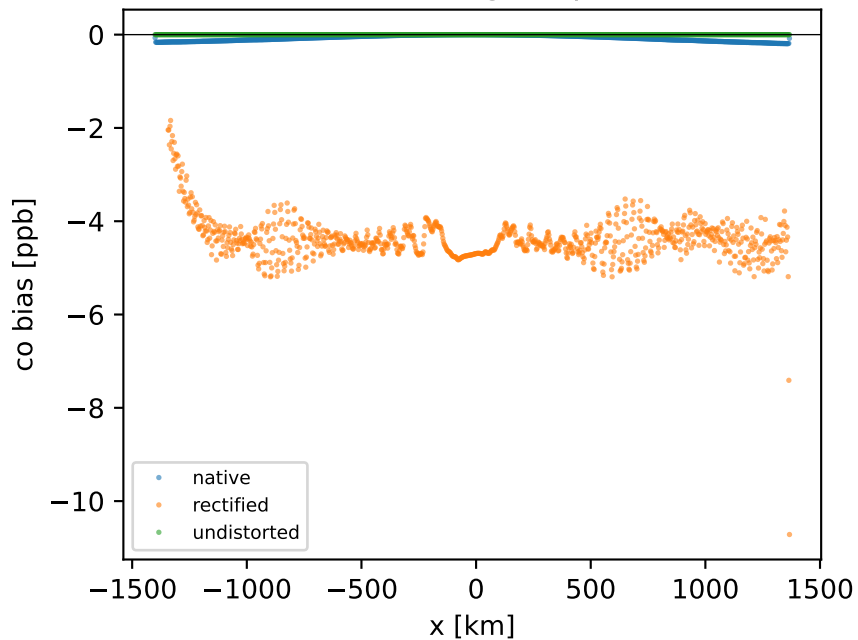


Joint FPA3 -- uniform, no noise (native/rectified/undistorted)

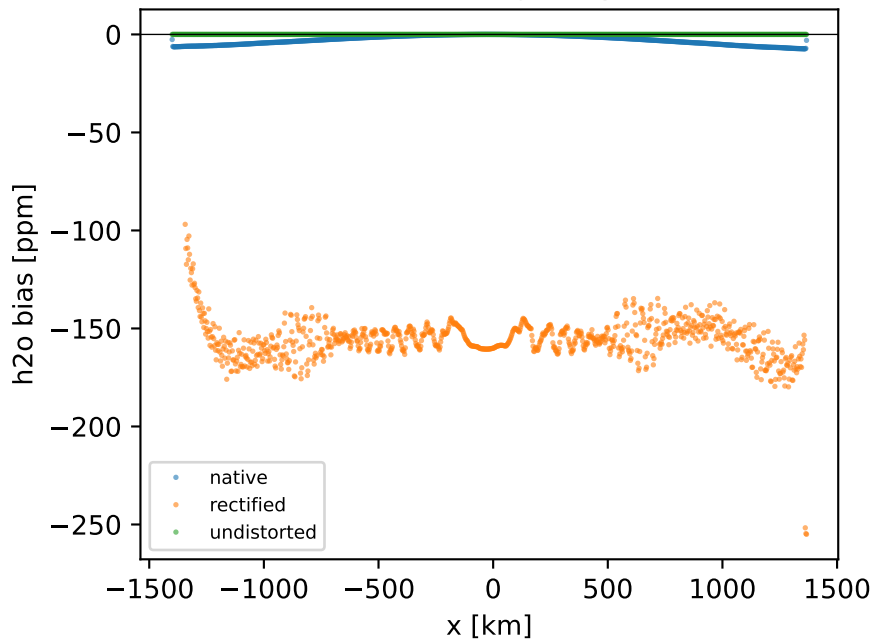
ch4 bias vs along-slit position



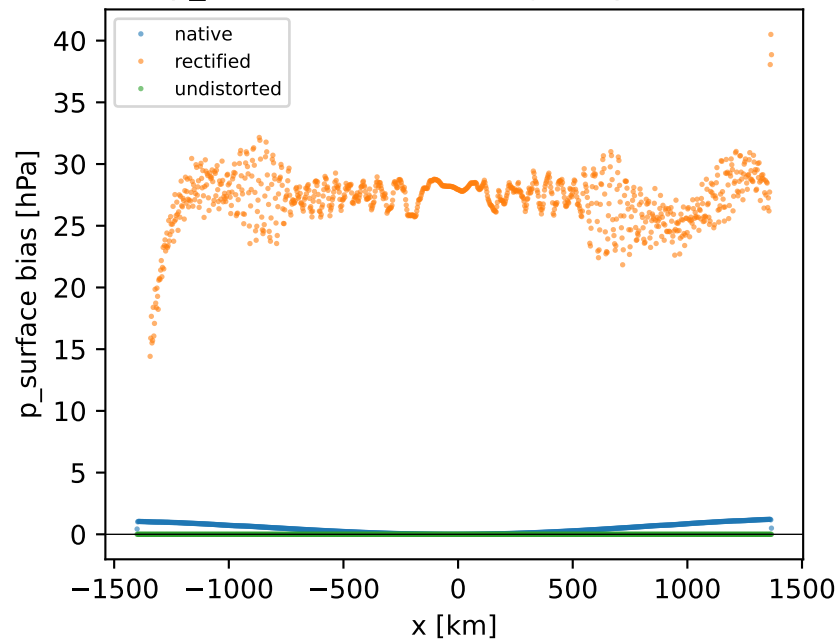
co bias vs along-slit position



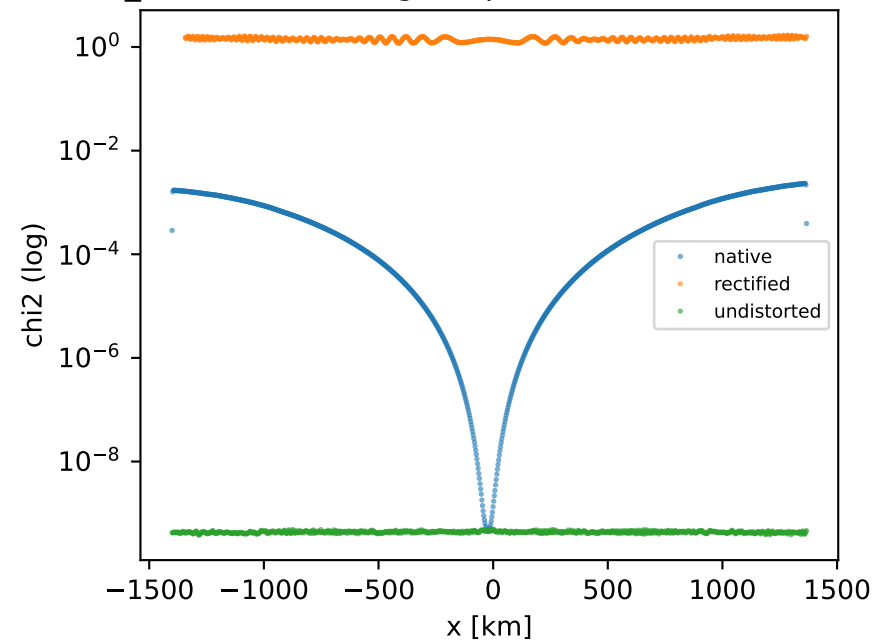
h2o bias vs along-slit position



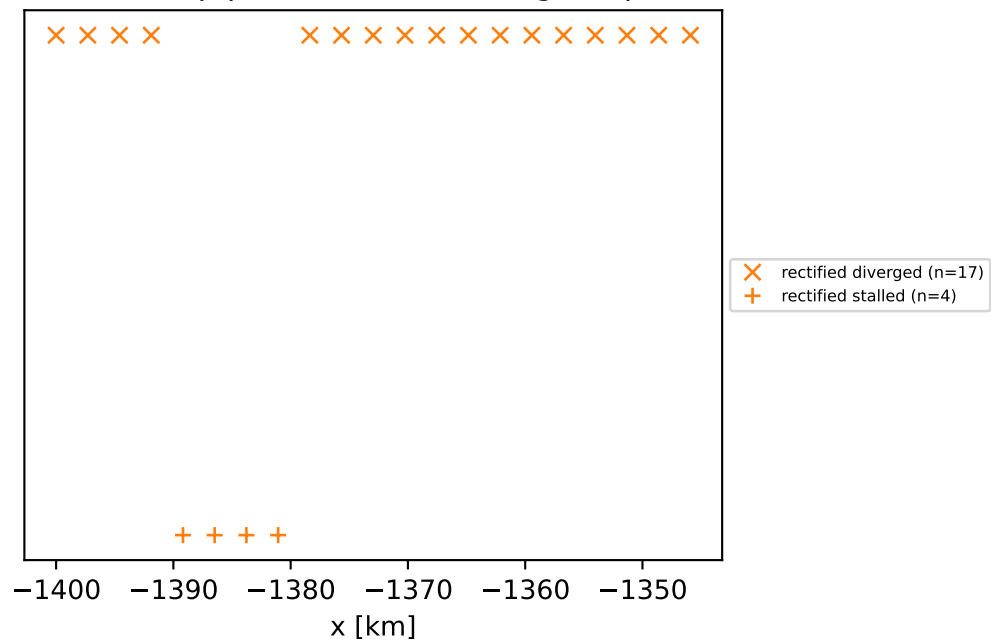
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

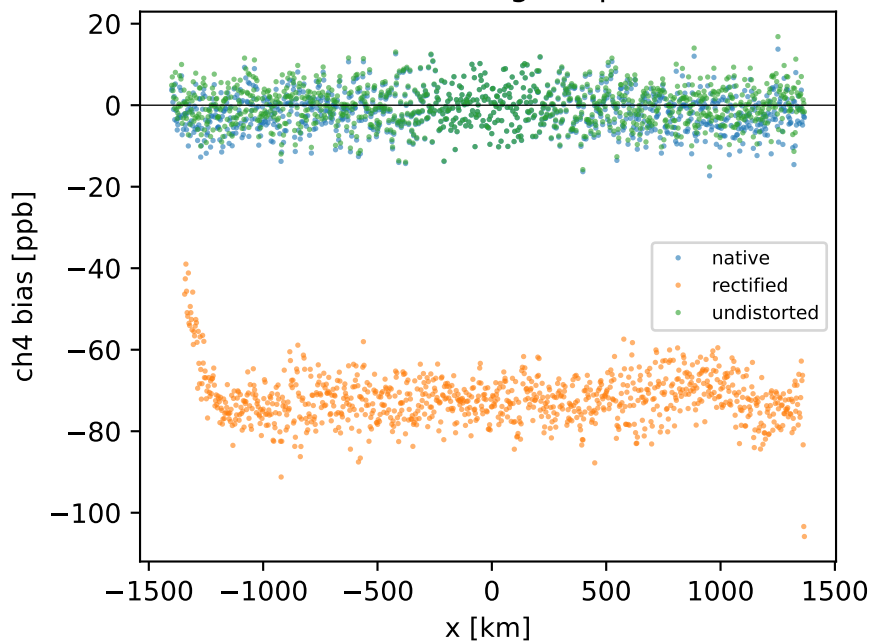


Where each pipeline fails, vs along-slit position

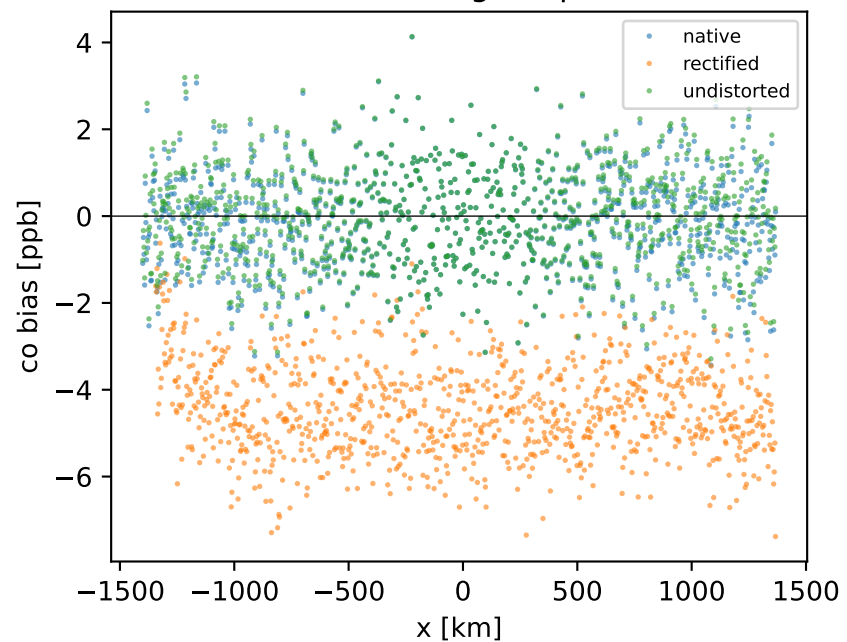


Joint FPA3 -- uniform, noise (native/rectified/undistorted)

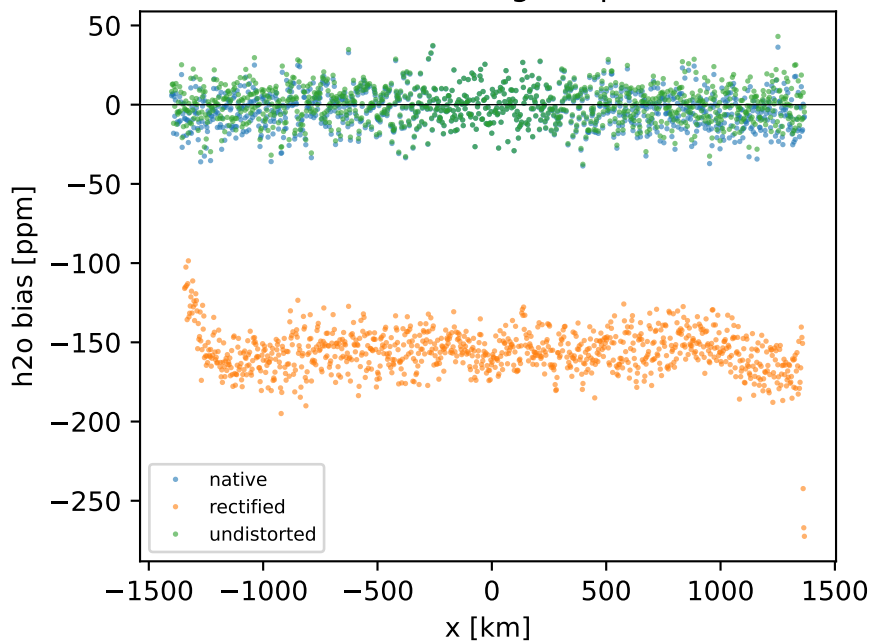
ch4 bias vs along-slit position



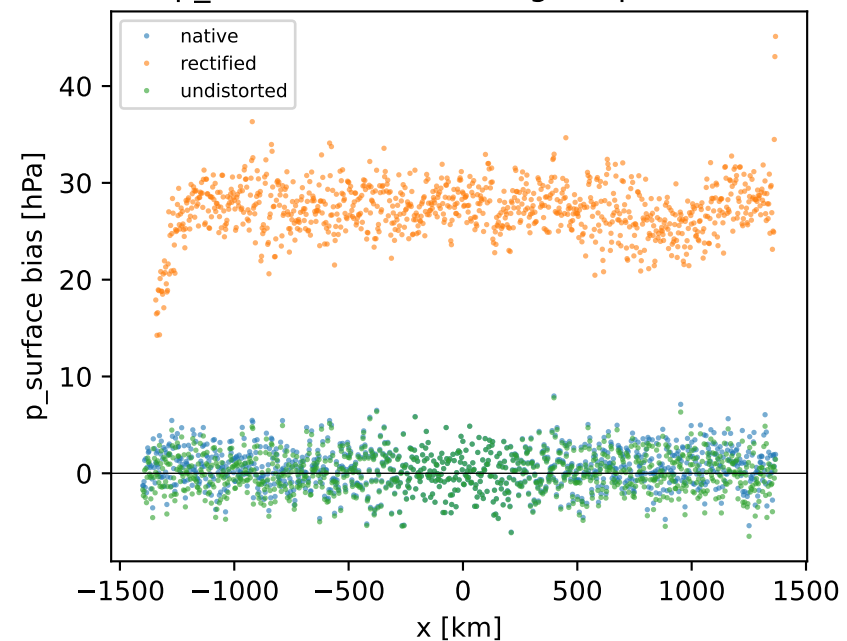
co bias vs along-slit position



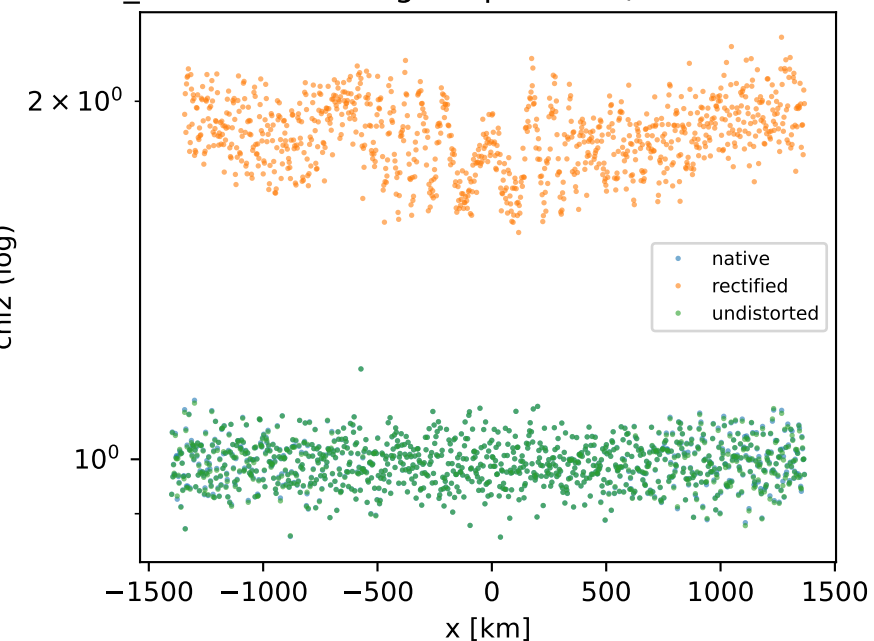
h2o bias vs along-slit position



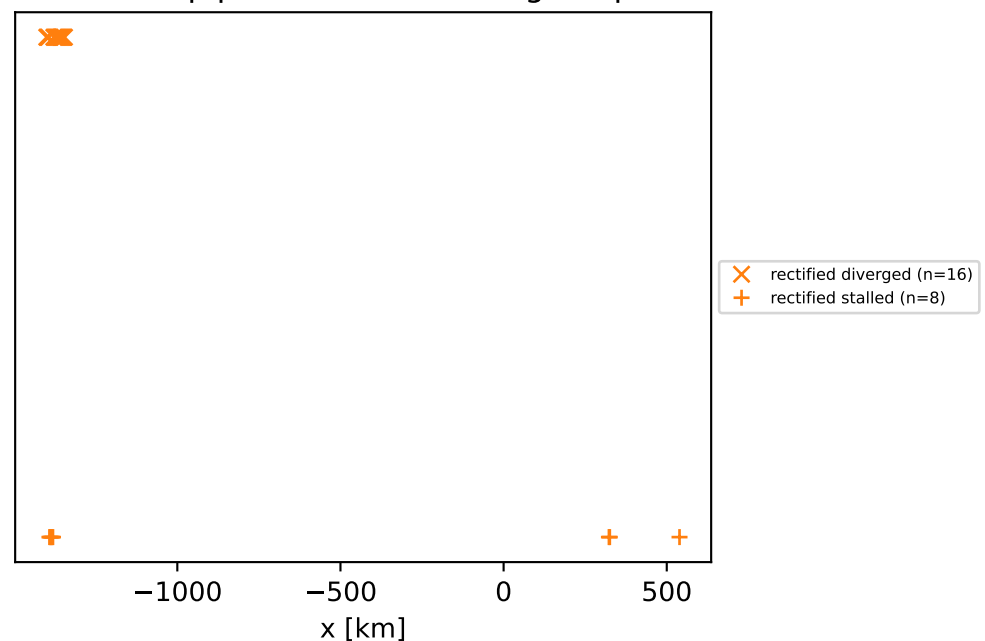
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

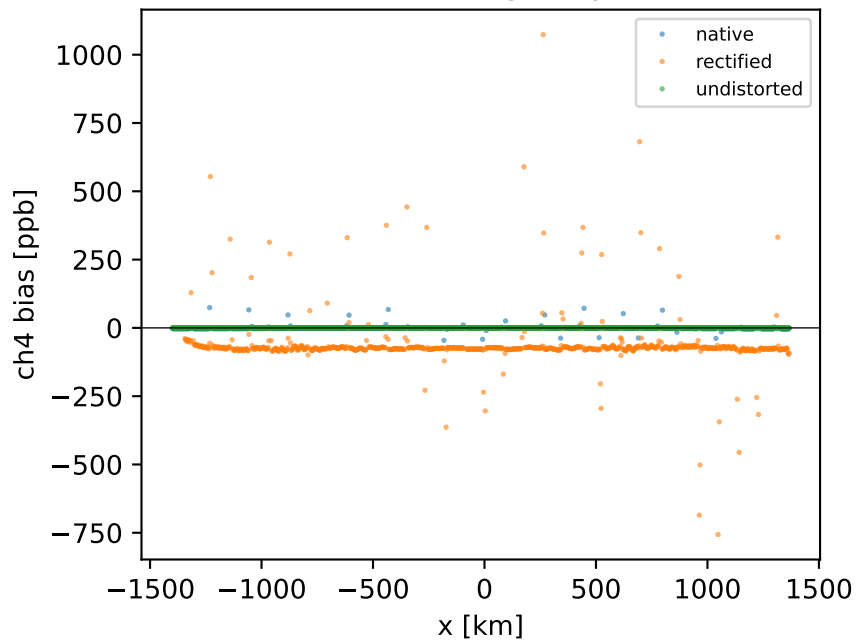


Where each pipeline fails, vs along-slit position

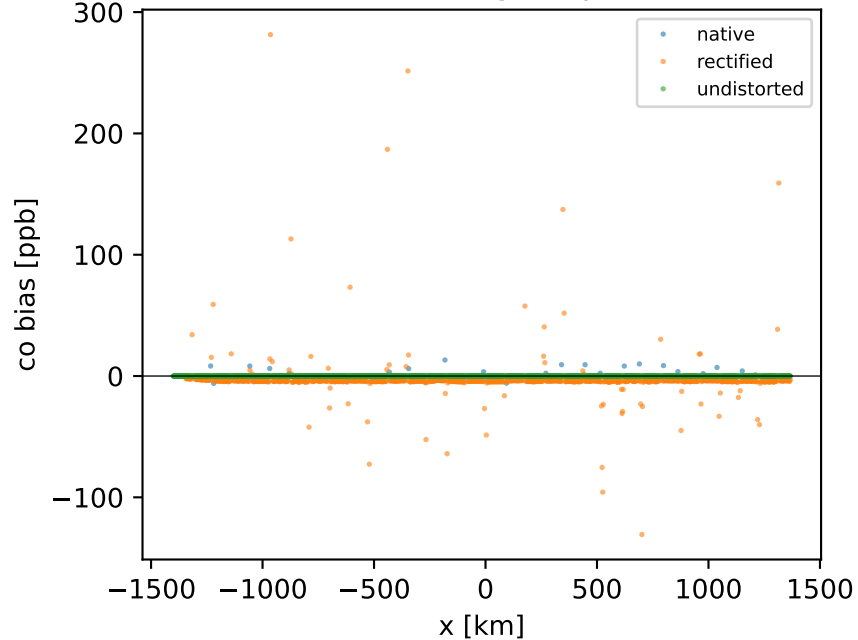


Joint FPA3 -- barcode, no noise (native/rectified/undistorted)

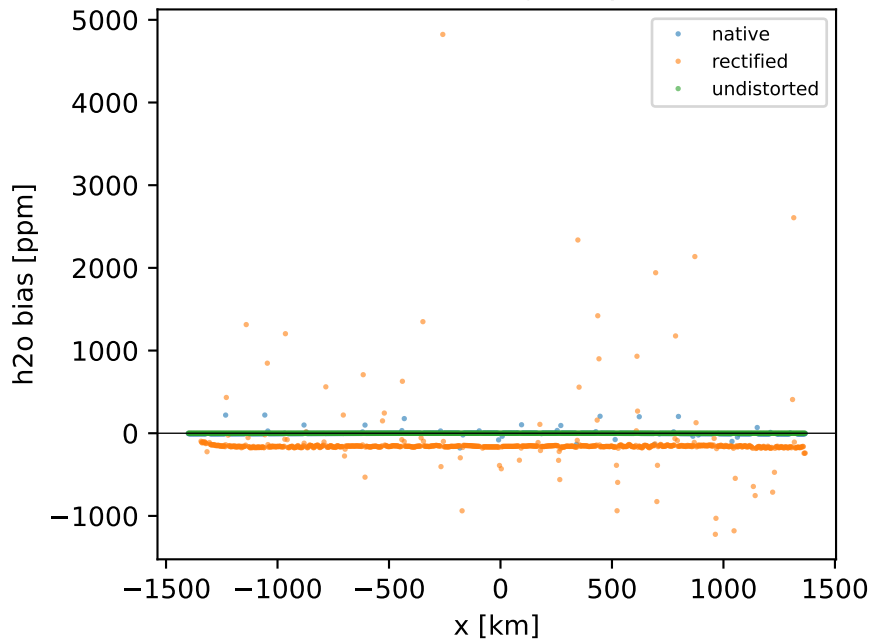
ch4 bias vs along-slit position



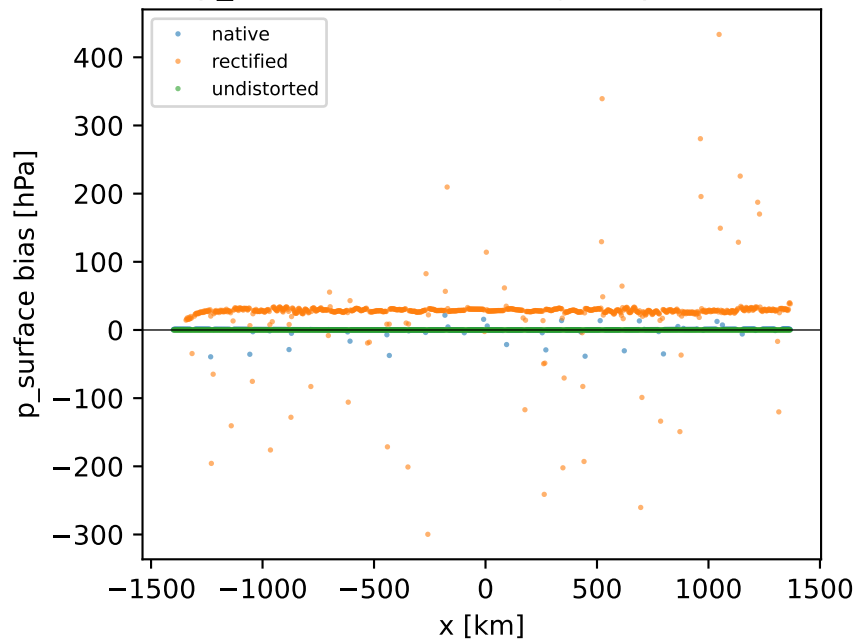
co bias vs along-slit position



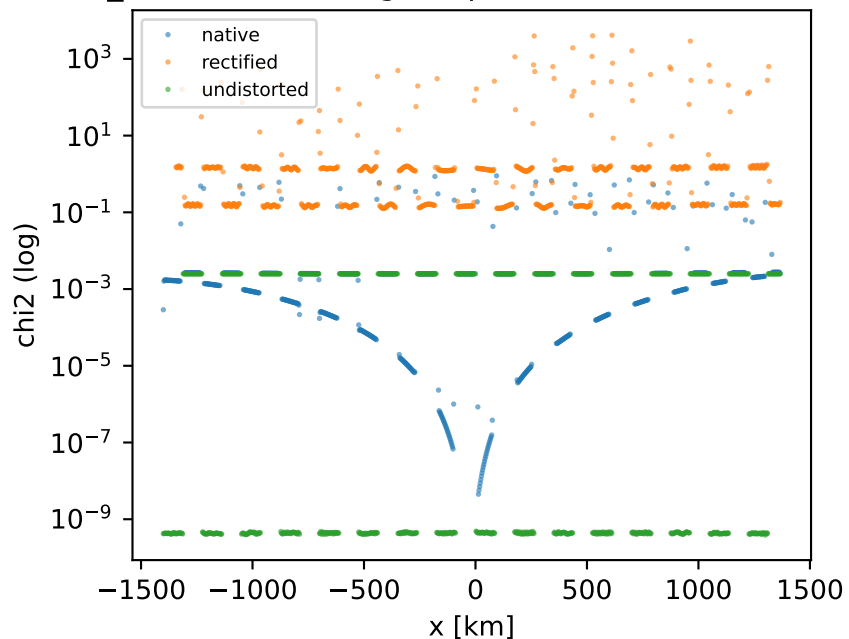
h2o bias vs along-slit position



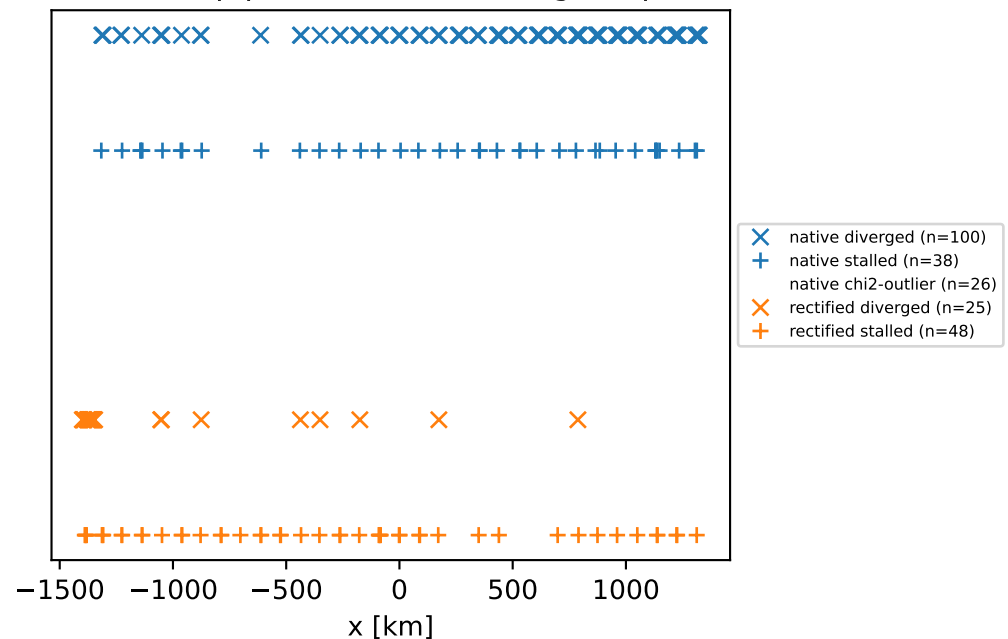
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

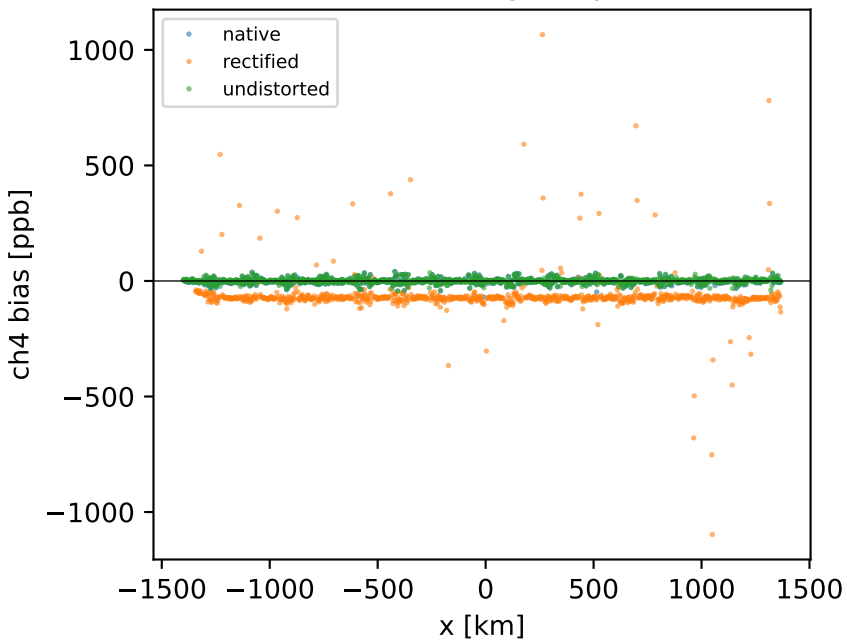


Where each pipeline fails, vs along-slit position

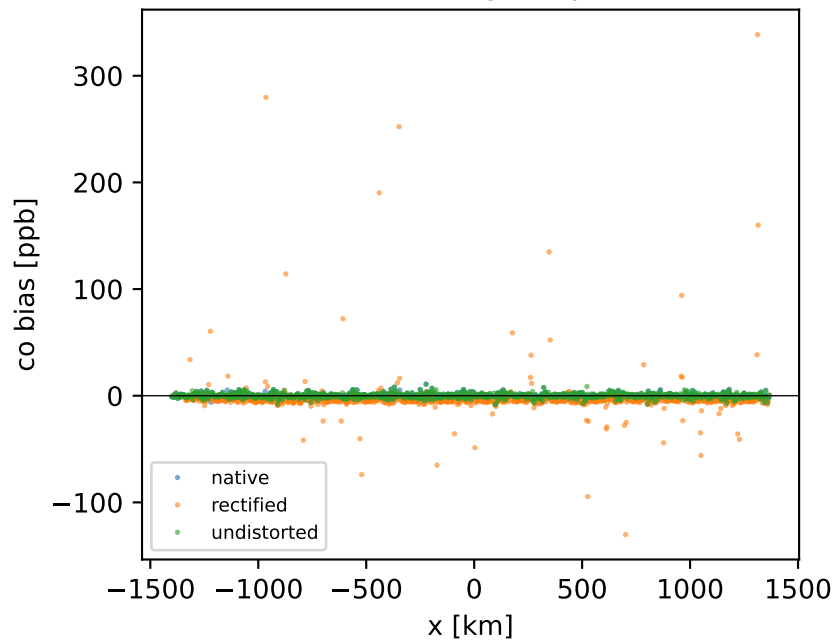


Joint FPA3 -- barcode, noise (native/rectified/undistorted)

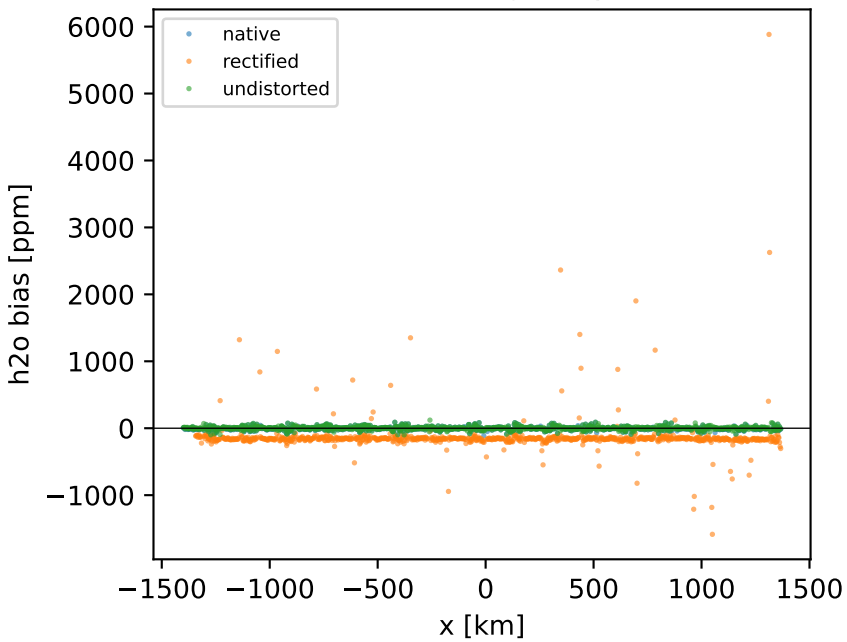
ch4 bias vs along-slit position



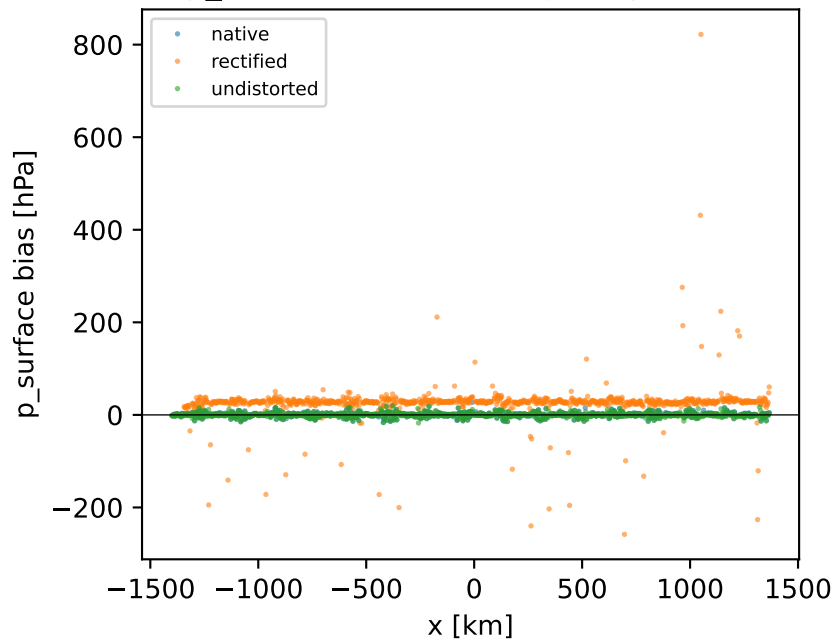
co bias vs along-slit position



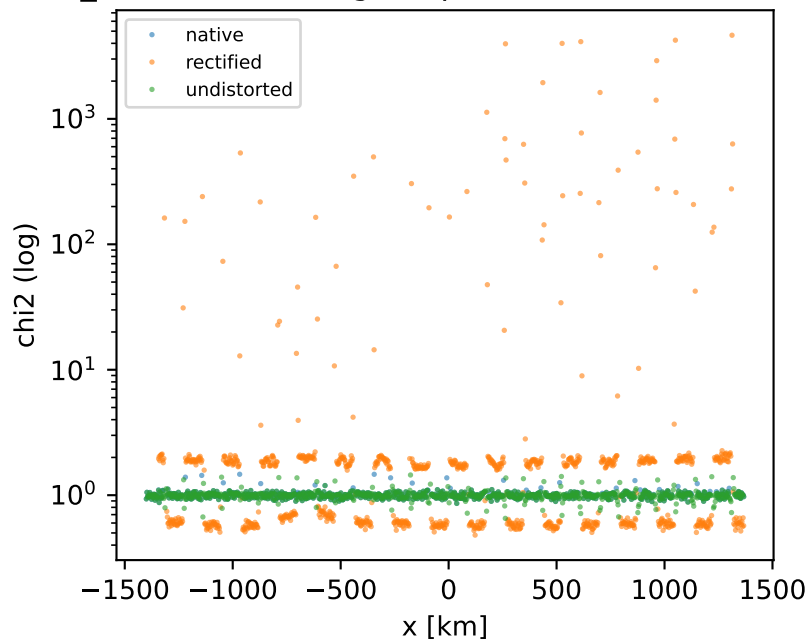
h2o bias vs along-slit position



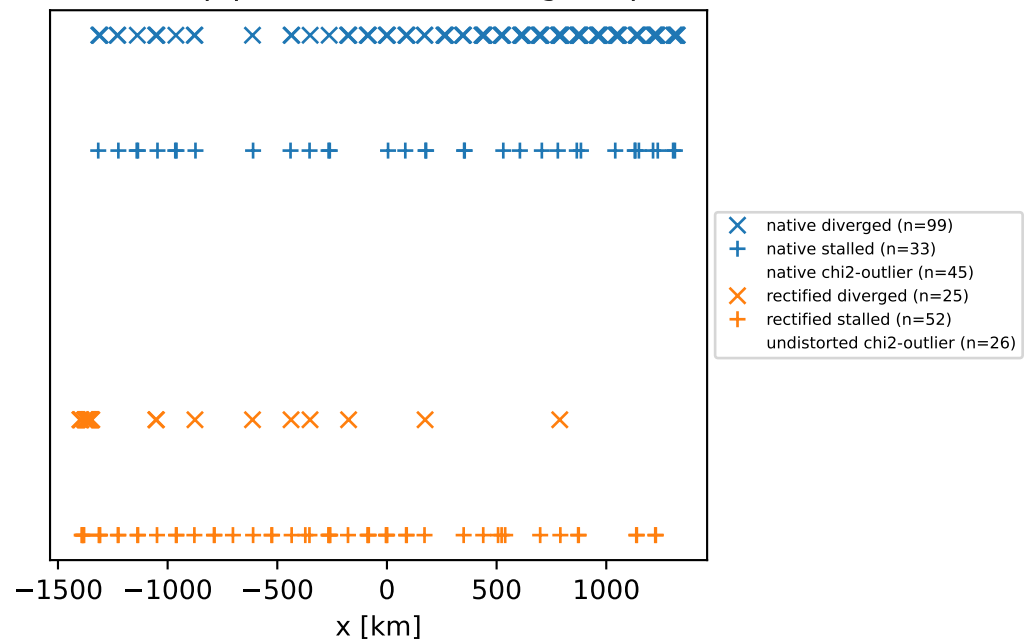
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)



Where each pipeline fails, vs along-slit position



FPA3 -- robust bias statistics across scenes/noise (native/rectified/undistorted) [page 1/2]

scene	noise	quantity	pipeline	n	mean	std	median	MAD-sig	n_out
realistic	0	ch4	native	1024	-2.161	3.345	-2.219	1.851	0
realistic	0	co	native	1024	-0.864	0.9825	-0.7479	0.2933	46
realistic	0	h2o	native	1024	+370.4	638.2	+283.8	898.7	0
realistic	0	p_surface	native	1024	+0.8432	1.537	+0.8353	0.8425	0
realistic	0	ch4	rectified	983	-76.95	6.412	-76.06	4.82	0
realistic	0	co	rectified	983	-6.215	2.213	-5.724	0.6538	46
realistic	0	h2o	rectified	983	+191.5	573.8	+197.4	618.8	0
realistic	0	p_surface	rectified	983	+29.93	2.35	+30.08	2.168	0
realistic	0	ch4	undistorted	1021	-1.227	2.042	-0.5929	0.5492	85
realistic	0	co	undistorted	1021	-0.7409	0.7537	-0.6266	0.3041	31
realistic	0	h2o	undistorted	1021	+379.9	637.3	+285.4	901.8	0
realistic	0	p_surface	undistorted	1021	+0.5673	0.9592	+0.2818	0.4055	32
realistic	1	ch4	native	1023	-2.205	6.151	-2.26	5.523	0
realistic	1	co	native	1023	-0.9045	1.535	-0.8485	1.289	0
realistic	1	h2o	native	1023	+371.3	638.1	+286.4	891.7	0
realistic	1	p_surface	native	1023	+0.8621	2.667	+0.8397	2.523	0
realistic	1	ch4	rectified	979	-77	7.576	-76.13	5.782	0
realistic	1	co	rectified	979	-6.244	2.415	-5.853	1.302	17
realistic	1	h2o	rectified	979	+193.7	572.9	+201.8	613.6	0
realistic	1	p_surface	rectified	979	+29.96	2.961	+30.08	2.826	0
realistic	1	ch4	undistorted	1023	-1.268	5.539	-1.079	5.124	0
realistic	1	co	undistorted	1023	-0.7807	1.397	-0.7145	1.267	0
realistic	1	h2o	undistorted	1023	+376.8	640.7	+287.8	898.5	0
realistic	1	p_surface	undistorted	1023	+0.5861	2.375	+0.4103	2.298	0
uniform	0	ch4	native	1024	-1.347	1.032	-1.232	1.406	0
uniform	0	co	native	1024	-0.07809	0.05894	-0.0726	0.0811	0
uniform	0	h2o	native	1024	-2.93	2.268	-2.652	3.063	0
uniform	0	p_surface	native	1024	+0.49	0.3752	+0.4489	0.5104	0
uniform	0	ch4	rectified	1003	-71.83	4.847	-72.55	2.871	10
uniform	0	co	rectified	1003	-4.38	0.4766	-4.423	0.3102	3
uniform	0	h2o	rectified	1003	-155.6	10.79	-155.8	7.089	4
uniform	0	p_surface	rectified	1003	+27.17	2.17	+27.47	1.489	3
uniform	0	ch4	undistorted	1024	-0.0005764	5.747e-05	-0.0005739	5.508e-05	0
uniform	0	co	undistorted	1024	-3.307e-05	1.037e-05	-3.367e-05	1.087e-05	0
uniform	0	h2o	undistorted	1024	-0.001567	0.000189	-0.001557	0.0001932	0
uniform	0	p_surface	undistorted	1024	+0.0001414	2.508e-05	+0.00014	2.296e-05	0
uniform	1	ch4	native	1024	-1.427	4.892	-1.428	4.918	0
uniform	1	co	native	1024	-0.1241	1.126	-0.1004	1.19	0
uniform	1	h2o	native	1024	-3.105	11.76	-2.844	11.71	0
uniform	1	p_surface	native	1024	+0.5302	2.07	+0.5112	2.002	0
uniform	1	ch4	rectified	1000	-71.93	6.129	-72.15	5.113	0
uniform	1	co	rectified	1000	-4.421	1.057	-4.501	1.05	0
uniform	1	h2o	rectified	1000	-155.9	13.83	-156.1	11.85	2
uniform	1	p_surface	rectified	1000	+27.22	2.77	+27.33	2.401	0

FPA3 -- robust bias statistics across scenes/noise (native/rectified/undistorted) [page 2/2]

uniform	1	ch4	undistorted	1024	-0.0806	4.818	+0.0334	4.743	0
uniform	1	co	undistorted	1024	-0.04609	1.125	-0.01958	1.19	0
uniform	1	h2o	undistorted	1024	-0.1752	11.55	-0.05963	11.78	0
uniform	1	p_surface	undistorted	1024	+0.04016	2.046	-0.02271	1.988	0
barcode	0	ch4	native	860	-0.873	7.326	-1.238	1.431	20
barcode	0	co	native	860	+0.01602	1.127	-0.06645	0.07895	38
barcode	0	h2o	native	860	-1.568	21.72	-2.597	3.012	25
barcode	0	p_surface	native	860	+0.1976	3.784	+0.4649	0.5402	24
barcode	0	ch4	rectified	951	-66.43	86.4	-74.12	5.033	52
barcode	0	co	rectified	951	-3.241	18.53	-4.191	0.4678	72
barcode	0	h2o	rectified	951	-134.6	276.2	-158.1	9.33	63
barcode	0	p_surface	rectified	951	+26.01	37.64	+28.43	2.443	56
barcode	0	ch4	undistorted	1024	-0.0003743	0.0002206	-0.0004638	0.0002883	0
barcode	0	co	undistorted	1024	+5.154e-05	8.64e-05	-9.542e-06	6.717e-05	0
barcode	0	h2o	undistorted	1024	-0.00152	0.0002513	-0.001533	0.0002216	1
barcode	0	p_surface	undistorted	1024	+0.000168	4.364e-05	+0.0001611	4.57e-05	0
barcode	1	ch4	native	847	-1.339	11.69	-1.407	7.92	1
barcode	1	co	native	847	-0.1469	2.212	-0.1961	1.833	0
barcode	1	h2o	native	847	-2.835	27.25	-2.295	18.36	0
barcode	1	p_surface	native	847	+0.523	4.966	+0.4257	3.191	1
barcode	1	ch4	rectified	947	-66.85	95.09	-73.43	8.421	45
barcode	1	co	rectified	947	-2.726	21.84	-4.415	1.737	48
barcode	1	h2o	rectified	947	-136.3	291.5	-157.6	20.81	47
barcode	1	p_surface	rectified	947	+26.85	43.75	+28.09	3.809	47
barcode	1	ch4	undistorted	998	-0.05549	11.31	+0.03339	7.703	0
barcode	1	co	undistorted	998	-0.02063	2.155	-0.03565	1.765	0
barcode	1	h2o	undistorted	998	-0.08048	26.77	+0.03249	17.25	0
barcode	1	p_surface	undistorted	998	+0.073	4.841	-0.04285	3.139	0